FIITJEE

JEE(Main)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

PART TEST – III
(Main)

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**B**	31.	**B**	61.	**D**
2.	**A**	32.	**B**	62.	**C**
3.	**D**	33.	**B**	63.	**B**
4.	**A**	34.	**A**	64.	**B**
5.	**A**	35.	**B**	65.	**B**
6.	**C**	36.	**C**	66.	**C**
7.	**B**	37.	**B**	67.	**A**
8.	**C**	38.	**B**	68.	**B**
9.	**C**	39.	**B**	69.	**A**
10.	**C**	40.	**C**	70.	**B**
11.	**B**	41.	**C**	71.	**B**
12.	**C**	42.	**C**	72.	**D**
13.	**A**	43.	**B**	73.	**A**
14.	**A**	44.	**B**	74.	**B**
15.	**A**	45.	**A**	75.	**C**
16.	**C**	46.	**D**	76.	**B**
17.	**A**	47.	**B**	77.	**C**
18.	**B**	48.	**A**	78.	**C**
19.	**A**	49.	**D**	79.	**C**
20.	**C**	50.	**C**	80.	**B**
21.	**C**	51.	**C**	81.	**B**
22.	**B**	52.	**D**	82.	**D**
23.	**C**	53.	**C**	83.	**A**
24.	**A**	54.	**A**	84.	**B**
25.	**B**	55.	**C**	85.	**A**
26.	**C**	56.	**C**	86.	**C**
27.	**A**	57.	**A**	87.	**C**
28.	**A**	58.	**C**	88.	**D**
29.	**A**	59.	**A**	89.	**C**
30.	**C**	60.	**A**	90.	**C**

Physics

PART – I

1. $A = A_0 e^{-b/2mt}$ $b = 6\pi r\eta$

$$\frac{A_0}{2} = A_0 e^{-b/2m.\Delta t}$$

$$\ell n2 = \frac{6\pi r\eta}{2m}.n$$

$$\eta = \frac{m\ell n2}{3\pi rn}$$

2. Positive half cycle
V_{output} = 10V
For negative half cycle.
potential difference across output = – 25 V

3. The distance by which image appears to be raised is

$$d = t_1\left(1-\frac{1}{\mu_1}\right)+t_2\left(1-\frac{1}{\mu_2}\right)$$

$$\frac{t}{2} = \frac{t}{2}\left(1-\frac{1}{\mu}\right)+\frac{t}{2}\left(1-\frac{2}{3\mu}\right)$$

$$\therefore \quad \mu = \frac{5}{3} = 1.67$$

4 Leftward Force = $\frac{IA}{c} = \frac{\Delta P}{\Delta t}$

5. The wavelength λ_{min} emitted by the X-ray tube operating at a voltage V is given by

$$eV = \frac{hc}{\lambda_{min}}$$

Kinetic energy of the electron = $\frac{1}{2}mv^2 = \frac{1}{2}\left(\frac{m^2v^2}{m}\right) = \frac{h^2}{2m\lambda_1^2}$

Now the wavelength λ of an electron moving with the velocity v is given by

$$\lambda = h/mv \Rightarrow \lambda = \frac{h}{\sqrt{2meV}}$$

$\therefore$ K.E. = 600 eV

6. $T = 2\pi\sqrt{\frac{\mu}{k}}$

$$T = \frac{2\pi}{\pi} \Rightarrow T = 2\,\text{sec}$$

Required time $= \frac{T}{4}+\frac{T}{4} = \frac{T}{2} = 1\text{sec}.$

7. I_0 is intensity of unpolarized incident light.

$I = \frac{I_0}{2}\cos^2 30^o \cos^2 60^o = 3$ watt m^{-2}.

8. $\lambda = \dfrac{h}{mv}$

9. $f = \dfrac{c}{\lambda} \Rightarrow f_m = \dfrac{c}{\lambda_m}$ & $f_s = \dfrac{c}{\lambda_s}$

$\because \quad \lambda_m > \lambda_s \Rightarrow f_m < f_s$

$f_m = \dfrac{1}{2\pi\sqrt{L_m c}}$ & $f_s = \dfrac{1}{2\pi\sqrt{L_s c}}$

$\Rightarrow \quad L_s < L_m$.

10. $v = \sqrt{\dfrac{T}{\mu}}$

T can be calculated by using Hooke's Law and on stretching μ also changes.

11. $\dfrac{T}{A} = \dfrac{Y\Delta\ell}{\ell}$

12. Here

$5\dfrac{\lambda}{4} = 42 + 0.3\times 10 \Rightarrow \lambda = 36\text{cm}$

$\therefore$ pressure amplitude at middle of pipe is

$\Delta P = \Delta P_o \left|\sin Kx\right| = \Delta P_o \left|\sin\left(\dfrac{2\pi}{36}\times 24\right)\right|$

$= \dfrac{\sqrt{3}\Delta P_o}{2}$

13. $d_{min} = \dfrac{1.22\lambda}{2\sin\beta}$

Where β is the angle subtended by the objective at the object.
For light of 5500 Å

$d_{min} = \dfrac{1.22\times 5.5\times 10^{-7}}{2\sin\beta}\text{m}$

For electrons accelerated through 100V the deBroglie wavelength is

$\lambda = \dfrac{h}{p} = \dfrac{1.227}{\sqrt{100}} = 0.13\times 10^{-9}\text{m}$

$\therefore \quad d'_{min} = \dfrac{1.22\times 1.3\times 10^{-10}}{2\sin\beta}$

$\therefore \quad d'_{min} = \dfrac{1.22\times 1.3\times 10^{-10}}{2\sin\beta}$

$\dfrac{d'_{min}}{d_{min}} = \dfrac{1.3\times 10^{-10}}{5.5\times 10^{-7}} \sim 0.2\times 10^{-3}$

14. $E = \Delta mc^2$

16. For 1st reading of oscillator
$f_A = (514 \pm 2)$Hz
$\Rightarrow$ f_A = 516 Hz or 512 Hz
For 2nd reading of oscillator
$f_A = (510 \pm 6)$ Hz
$\Rightarrow$ f_A = 516 Hz or 504 Hz
$\Rightarrow$ A has a frequency of 516 Hz

17. $$\frac{1.22\lambda}{d} = \theta$$
$$\frac{1.22 \times 5.4 \times 10^{-5}}{12} = \frac{30}{x}$$
$$x = 54.645\,\text{km}$$

18. Frequency received by car B is $f_1 = \left(\frac{v + w + u}{v + w - u}\right) f$
Now the car B will be treated as a source of frequency f_1
$\therefore$ wavelength of reflected sound received by the driver of car A is
$$\lambda' = \frac{(v - w - u)}{f_1} = \frac{(v - w - u)(v + w - u)}{(v + w + u) f}$$
$$\frac{(330 - 5 - 25)(330 + 5 - 25)}{(330 + 5 + 25)300} = \frac{31}{36}\text{m}$$

19. For continuity displacement current should be equal to conduction current
$$q = q_0\left(1 - e^{-t/RC}\right) \Rightarrow |i| = \frac{q_0}{RC} e^{-t/RC}\ CR = t$$
$$\Rightarrow \quad \frac{1200\,\mu\text{F}}{ex1} \Rightarrow i = \frac{0.0012}{e}\,\text{A} = \frac{1.2}{e}\,\text{mA}$$

20. $v_{max} = A\omega$

21. $$\rho = \frac{m}{\frac{4}{3}\pi r^3} \Rightarrow \frac{\Delta\rho}{\rho} = \frac{\Delta m}{m} + \frac{3\Delta r}{r}$$
$$\Rightarrow \quad \frac{\Delta\rho}{\rho} = \frac{0.001}{30} + \frac{3 \times 0.01}{10}$$
$$\frac{\Delta\rho}{\rho} = 3.3 \times 10^{-5} + 3 \times 10^{-3}$$
$$\frac{\Delta\rho}{\rho} = 0.033 \times 10^{-3} + 3 \times 10^{-3}$$
$$\frac{\Delta\rho}{\rho} = 3.033 \times 10^{-3}$$

22. $$2T\sin\theta = \Delta P \times \text{Area}$$
$$\left(\frac{T}{A}\right).\theta = \Delta P \Rightarrow \frac{\sigma_0 \Delta r}{r} = \Delta P$$

23. 2 minutes = 3 half life

1 half life = $\frac{120}{3} = 40$ sec

24. $f_0 + f_e = 40$

$7 = \frac{f_0}{f_e} \Rightarrow f_0 = 35\,cm$

$f_e = 5\,cm$

25. $T_1 = k(\ell_1 - \ell_0)$

$T_2 = k(\ell_2 - \ell_0)$

$\Rightarrow \quad \ell_0 = \frac{\ell_2 T_1 - \ell_1 T_2}{T_1 - T_2}$

26. $\frac{g}{2} = g - R\omega^2$

$\Rightarrow \quad \omega = \sqrt{\frac{g}{2R}}$

27. $P = \frac{1}{2}Q\rho\left(V^2 - V_0^2\right) + \rho Qgh$

$\sqrt{\frac{2P}{Q\rho} - 2gh + V_0^2} = V$

28. $\overline{A + B} = \overline{A} \cdot \overline{B}$

29. Buoyancy = effective weight of liquid displaced

30. $T = \pi\left[\sqrt{\frac{M}{K}} + \frac{1}{2}\sqrt{\frac{M}{K}}\right] = \frac{3\pi}{2}\sqrt{\frac{M}{K}}$

Chemistry

PART – II

31. Oxidation number of Mn in $\left[MnO_4\right]^-$ is 7 and Cr in CrO_2Cl_2 +6.

32. Volume of all atoms in 1.68 g of Ar

$$= \frac{1.68}{40} \times 6 \times 10^{23} \times \frac{4}{3} \times \frac{22}{7} \times \left(1.50 \times 10^{-8}\right)^3 = 0.356$$

% of empty space = (1 – 0.3564) × 100 = 64.36

33. $G^* = \frac{\ell}{A} = \frac{0.02}{0.0004} = 50\ m^{-1}$

$$\frac{1}{R} = \frac{1}{50}$$

Specific conductance $= G^* \times \frac{1}{R} = 1\ S\,cm^2$

34. When impure metal has impurity of its own metal oxide, then poling process is used e.g. impure Cu and Sn are purified by this method.

35. Equivalent weight of $KBrO_3$ $\frac{167}{6(n-\text{factor})}$ n-factor = 6

$$N_{Na_2S_2O_3} = \frac{0.167}{167} \times 6 \times \frac{1}{0.05} = 0.12\ N$$

36. Mass of 1 litre of solution = 1010 gram

Mass of solvent = 1010 – 300 × 0.1 = 980 gram

$$m = \frac{0.1}{0.98} = 0.102$$

$$\Delta T_f = iK_f m = (1+\alpha)K_f.m$$

$$\Rightarrow \Delta T_f = 0.199^oC$$

$$\Rightarrow T_f = 0 - 0.199^oC = -0.199^oC$$

37. Effective number of O^{2-} in a unit cell $= 8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$

Effective number of A^{2+} in a unit cell = $8 \times \frac{1}{8} = 1$

Effective number of B^{3+} in a unit cell = $4 \times \frac{1}{2} = 2$

$\therefore$ General formula = AB_2O_4

38. $\underset{\text{Impure}}{Ni} + 4CO \longrightarrow \underset{\text{Volatile compound}}{\left[Ni(CO)_4\right]} \uparrow \xrightarrow[\text{Thermal decomposition}]{200 \text{ to } 300^oC} \underset{\text{Pure}}{Ni} + 4CO \uparrow$

39. $\underset{\text{(Black)}}{FeS} \xrightarrow{\text{Dil. } H_2SO_4} H_2S \xrightarrow{\text{Dil. } HNO_3} \underset{\text{Colloidal sulphur}}{S} + NO_2 + 2H_2O$

40. At constant V $= \frac{P_1}{T_1} = \frac{P_2}{T_2} \Rightarrow P_2 = 1.5$ atm

For single phase $dG = VdP - SdT$

$\Rightarrow \Delta G = v.\Delta P - \int \left(2 + 10^{-2} T\right) dT$

= 781.5 J

41. $\sqrt{3}a = 2\left(r^+ + r^-\right)$

$\Rightarrow a = \frac{2 \times (1.69 + 1.81)}{1.732}$

$\Rightarrow a = 4.04 \overset{o}{A}$

42. For normal Calomel electrode $E_{RP} = E^o_{RP}$

$E_{cell} = (0.28 - 0) - \frac{0.059}{2} \log \frac{\left[H^+\right]^2}{P_{H_2}} \Rightarrow pH = 5.5$

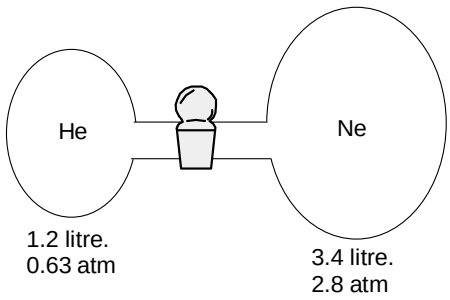
43. As $PV^\gamma = constant \Rightarrow p.\gamma v^{\gamma-1} \times dv + V^\gamma.dp = 0$

$\frac{dP}{P} = -\gamma . \frac{v^{\gamma-1}}{v^\gamma} dv$

$\Rightarrow \frac{dp}{p} = -\gamma \frac{dV}{V}$

44. Preferential adsorption of Fe^{3+} takes place so positive charge develop.

45.



$n_{He} = \frac{1.2 \times 0.63}{0.082 \times 289}$ = 0.032

$n_{Ne} = \frac{3.4 \times 2.8}{0.0821 \times 289}$ = 0.4

$P_T = \frac{0.432 \times 0.0821 \times 289}{4.6} = 2.223$

$P_{He} = \frac{0.032}{0.432} \times 2.223 = 0.164$ atm

46. $BaCl_2 + KI \longrightarrow$ No ppt.

$BaCl_2(aq) + CrO_4^{2-} \longrightarrow \underset{\text{Yellow ppt.}}{BaCrO_4} \downarrow$

47. $\Delta H, \Delta S$ & ΔG should all be – ve to hold chemical adsorption.

48. Eq. wt. of $N_2H_4 = \dfrac{\text{Molar mass}}{4}$

Eq. wt. of $IO_3^- = \dfrac{\text{Molar mass}}{4}$

49. $\left[M(AA)_2\right]^{n\pm}$ does not show geometrical isomerism and can not show optical.

50. $2CH_3OH(\ell) + 3O_2 \longrightarrow 4H_2O(\ell) + 2CO_2(g) + 1453$

$4 \times 44 + 4H_2O \longrightarrow 4H_2O$

$2CH_3OH(\ell) + 3O_2(g) \longrightarrow 4H_2O(\ell) + 2CO_2(g) + 1277$

$\Delta H = -1277$ kJ

51. The word smog is originated from smoke and fog. Hence it is oxide of sulphur and N_2.

52. Piperazine is not a tranquilizer.

53. As in higher oxidation state transition elements show acidic nature.

54. $\underset{\text{Fused anhydrous}}{MgCl_2} \longrightarrow Mg^{2+} + 2e^-$

At cathode: $Mg^{2+} + 2e^- \longrightarrow Mg$

At anode: $2Cl^- - 2e^- \longrightarrow Cl_2 \uparrow$

55. Phenelzine is a tranquilizer. It is not an antacid.

57. 1 watt = 1 J/sec
Total heat for 36 mL H_2O
= 806 × 100
= 80600 J

$\Delta H_{\text{vaporisation}} = \dfrac{80600}{36} \times 18$

= 40300 J or 40.3 kJ/mole

58. At anode,

$2HgO(\ell) \longrightarrow 4H^+(aq) + O_2(g) + 4e^-$

At cathode

$Ag^+(aq) + e^- \longrightarrow Ag(s)$

Ag deposited = 0.1 mole = 10.8 gram

O_2 expelled out = $\dfrac{0.1}{4} = 0.8$ gram

Total wt. loss of the solution = 11.6 gram
wt. of remaining solution = 108 – 11.6 = 96.4 gram

59. On applying valence bond theory
Ni has electronic configuration [Ar]$3d^8$ $4s^2$ and in presence of CO (strong field ligand), electronic configuration is $3d^{10}$.
Hybridization is sp^3 and all the electrons are paired.

Mathematics

PART – III

61. $x - y + 2 = 0$
$x + 2y = 4$
$x = 0, y = 2$
$\vec{a} = 2\hat{j} + 2\hat{k}$ and $|\vec{a}|^2 = 8$

$$\left[\vec{a}\ \vec{b}\ \vec{c}\right] = \begin{vmatrix} 0 & 2 & 2 \\ 1 & -1 & 1 \\ 1 & 2 & 0 \end{vmatrix} = 8 = |\vec{a}|^2$$

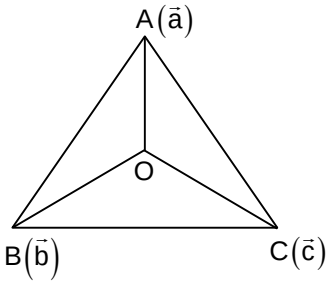
62. Vector $\left(2\hat{i} - 3\hat{j}\right) \times \left(\hat{i} - 4\hat{j} + 4\hat{k}\right)$ is perpendicular to $2\hat{i} - \hat{j} + m\hat{k}$

$$\Rightarrow \begin{vmatrix} 2 & -3 & 0 \\ 1 & -4 & 4 \\ 2 & -1 & m \end{vmatrix} = 0 \Rightarrow m = \frac{-16}{5}$$

63. Equation of line L is $\vec{r} = \left(\hat{j} - \hat{k}\right) + \lambda \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 4 \\ 3 & 2 & -2 \end{vmatrix}$

$\vec{r} = \left(\hat{j} - \hat{k}\right) + \lambda\left(-10\hat{i} + 16\hat{j} + \hat{k}\right)$
Point Q is $(-10\lambda, 16\lambda + 1, \lambda - 1)$
$\left|\overrightarrow{PQ}\right|^2 = 357\lambda^2 = 357$
$\lambda^2 = 1$
$\lambda = \pm 1$
Q ≡ (–10, 17, 0) or (10, –15, –2) then a + 2b + 3c = 24 or –26

64. Let $\overrightarrow{OA} = \vec{a}$; $\overrightarrow{OB} = \vec{b}$; $\overrightarrow{OC} = \vec{c}$



Area of $\Delta ABC = \frac{1}{2}\left|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}\right|$

Now, $\vec{r} = 3\vec{a} + 4\vec{b} + 6\vec{c} = 0$
$\vec{a} \times \vec{r} = 4\vec{a} \times \vec{b} + 6\vec{a} \times \vec{c} = 0$
$4\vec{a} \times \vec{b} = 6\vec{c} \times \vec{a} \Rightarrow \vec{c} \times \vec{a} = \frac{2}{3}\left(\vec{a} \times \vec{b}\right)$
$\vec{b} \times \vec{r} = 3\vec{b} \times \vec{a} + 6\vec{b} \times \vec{c} = 0$
$3\vec{a} \times \vec{b} = 6\vec{b} \times \vec{c} \Rightarrow \vec{b} \times \vec{c} = \frac{1}{2}\vec{a} \times \vec{b}$ and Area of Δ AOB is $\frac{1}{2}\left|\vec{a} \times \vec{b}\right|$

$$\text{Now, } \frac{\text{Area of } \Delta ABC}{\text{Area of } \Delta AOB} = \frac{\frac{1}{2}\left|\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}\right|}{\frac{1}{2}\left|\vec{a} \times \vec{b}\right|}$$

$$= \frac{\frac{1}{2}\left|\vec{a} \times \vec{b}\right|\left|1 + \frac{1}{2} + \frac{2}{3}\right|}{\frac{1}{2}\left|\vec{a} \times \vec{b}\right|} = \frac{13}{6}$$

65. $\left(\hat{a}\times\hat{b}\right)\times\left(\hat{a}+\hat{b}\right)=\left(1+\hat{a}\cdot\hat{b}\right)\left(\hat{b}-\hat{a}\right)$

66. $v_1=\left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right]$

$v_2=\left[\bar{a}+2\bar{b}-\bar{c}\quad \bar{a}-2\bar{b}\ 2\bar{c}\quad \bar{a}-4\bar{b}+2\bar{c}\right]=18\left[\bar{a}\ \ \bar{b}\ \ \bar{c}\right]$

67. $\sum_{k=0}^{100}\frac{k+1-1}{k+1}{}^{100}C_k=\sum_{k=0}^{100}{}^{100}C_k-\sum_{k=0}^{100}\frac{{}^{100}C_k}{k+1}=2^{100}-\frac{1}{101}\sum_{k=0}^{100}\frac{101}{k+1}{}^{100}C_k$

$=\frac{99\cdot 2^{100}+1}{101}$ (x = 99, y = 1 and z = 101) $\Rightarrow xy^2z = 9999$

68.

Tens place (A)	Unit place (B)
1, 2, 3 not possible	
4	0
5	0, 1
6	0, 1, 2
7	0, 1, 2, 3
8	0, 1, 2, 3, 4
9	0, 1, 2, 3, 4, 5

$P=\frac{21}{90}=\frac{7}{30}$

69. $D = -(\alpha + 2)(\alpha - 1)^2$

$D_1 = -(\alpha - 1)^2(\alpha + 2)$

$D_2 = D_3 = 0$

From $\alpha = 1, -2$ infinite solution

$\alpha \neq 1, -2$ unique solution

So, $L = 11; M = 0; N = L; L^2 + M^2 + N^2 = 125$

70. $\alpha+\beta+\gamma=\frac{5}{3}\ \Rightarrow\ \frac{\alpha+\beta+\gamma}{3}=\frac{5}{9}$

71. ${}^{22}C_2 + {}^{21}C_2 + {}^{20}C_2 + {}^{19}C_2 + ... + {}^{3}C_2 + {}^{2}C_2 = {}^{23}C_3$

72. $\sum_{r=1}^{n}a_r z^r=1$

$a_1z + a_2z^2 + ... + a_nz^n = 1$

$|a_1z + a2z^2 + ... + a_nz^n| = 1$

$1 \le |a_1||z| + |a_2||z|^2 + ... + |a_n||z^n| < 2[|z| + |z|^2 + ... + |z|^n \ \ \because\ |a_r| < 2$

$1\le 2\left[\frac{|z|\left[1-|z|^n\right]}{1-|z|}\right]$

$1 - |z| < 2|z| - 2|z|^{n+1}$

$2|z|^{n+1} - 3|z| + 1 < 0$

$|z|>\frac{1}{3}+\frac{2}{3}|z|^{n+1}$

$|z|>\frac{1}{3}$ which is contradiction

73. $A_k = e^{i\frac{k\pi}{20}} \Rightarrow \dfrac{\sum_{k=0}^{2018}\left|e^{i\frac{k\pi}{20}}\right|\left|e^{\frac{i\pi}{20}}-1\right|}{\sum_{k=1}^{673}\left|e^{i\frac{k\pi}{20}}\right|\left|1-e^{\frac{i\pi}{20}}\right|} = 3$

74. $\Delta = \dfrac{1}{\prod(a-p)^2}\begin{vmatrix} 1 & a-p & \dfrac{(a-p)^2}{p} \\ 1 & a-q & \dfrac{(a-q)^2}{q} \\ 1 & a-r & \dfrac{(a-r)^2}{r} \end{vmatrix}$

Apply $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$ and then on explained, we get $\Delta = \dfrac{-a^2\prod(p-q)}{prq\prod(a-p)^2}$

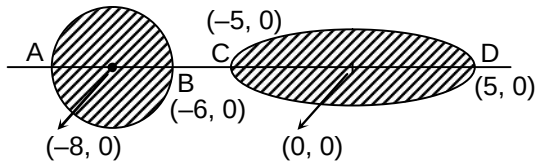
75. $A = \sum(a_r + b_r + c_r) = \sum a_r + \sum b_r + \sum c_r$

$\left(\sum a_r \cdot \sum b_r \cdot \sum c_r\right)^{1/3} \le \dfrac{\sum a_r + \sum b_r + \sum c_r}{3}$

$\sum a_r \cdot \sum b_r \cdot \sum c_r \le \dfrac{A^3}{27}$

$$\Delta = \begin{vmatrix} \sum a_r & a_2 & a_3 \\ \sum b_r & b_2 & b_3 \\ \sum c_r & c_2 & c_3 \end{vmatrix} = \sum a_r \sum b_r \sum c_r \begin{vmatrix} 1 & \dfrac{a_2}{\sum a_r} & \dfrac{a_3}{\sum a_r} \\ 1 & \dfrac{b_2}{\sum b_r} & \dfrac{b_3}{\sum b_r} \\ 1 & \dfrac{c_2}{\sum c_r} & \dfrac{c_3}{\sum c_r} \end{vmatrix} \le \sum a_r \sum b_r \sum c_r \le \frac{A^3}{27}$$

76. AD is maximum distance
BC is minimum distance
Range is [1, 15]



77. Area of the triangle formed by z, iz and z + iz is $\dfrac{1}{2}|z|^2$

78. $k_1 = 4! \times 5!$

$k_2 = \dfrac{8!}{5!}$

$\dfrac{7k_1}{k_2} = 60$

79. ${}^{(100-55)+(10-1)}C_{10-1} = {}^{54}C_9$

80. Letters C must be in middle of word
Required number are $\frac{6!}{2!\times 2!}$

81. Required probability is $\frac{1}{13}$

82. $\frac{5!\times 2!}{7!} = \frac{1}{{}^{7}C_{2}}$

83. $I_n + I_{n+2} = \frac{1}{n+1}$
Hence, common difference = 1

84. $t_r = \frac{8r}{4r^4+1} = \frac{8r}{\left(2r^2+1+2r\right)\left(2r^2+1-2r\right)}$
$S_n = 2\left[1-\frac{1}{2n^2+1+2n}\right]$ by method of difference
Hence, $S_{20} = \frac{1680}{841}$

85. $\frac{G_1G_2}{H_1H_2} = \frac{A_1+A_2}{H_1+H_2}$ in this case

86. $P(A \cap B) = P(A)\cdot P\left(\frac{B}{A}\right) = (0.4)(0.5) = 0.2$
Now $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.6 - 0.2 = 0.8$
Also, $P\left(\frac{\bar{A}}{\bar{B}}\right) = P\left(\frac{\bar{A}\cap\bar{B}}{P(\bar{B})}\right) = \frac{1-P(A\cup B)}{1-P(B)} = \frac{0.2}{0.4} = \frac{1}{2}$

87. Given f(f(–1)) = 0 and f(f(1)) = 0
i.e., f(f(x)) = 0 has of two roots f(1) and f(2)

88. Let ω_i (i = 1, 2, 3, 4, 5) denotes 1, 2, 3, 4, 5 white balls in the bag.
$P\left(\frac{\omega}{\omega_i}\right) = \frac{i}{5}$ (i = 1, 2, 3, 4, 5)
$P(\omega_i) = \frac{1}{5}$ (i = 1, 2, 3, 4, 5)
Now, $P\left(\frac{\omega}{\omega_5}\right) = \frac{P(\omega_5)P\left(\frac{\omega}{\omega_5}\right)}{\sum_{i=1}^{5} P(\omega_i)P\left(\frac{\omega}{\omega_i}\right)}$
$= \frac{\frac{1}{5}\times 1}{\frac{1}{5}\left(\frac{1}{5}+\frac{2}{5}+\frac{3}{5}+\frac{4}{5}+\frac{5}{5}\right)} = \frac{5}{15} = \frac{1}{3}$

89. $x_1, x_2 \ldots\ldots x_n$ be n observations then, $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$

Let $y_i = \frac{x_i}{\alpha} + 10$ then $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i = \frac{1}{\alpha}\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right) + \frac{1}{n}(10n)$

$\Rightarrow \bar{y} = \frac{\bar{x}}{\alpha} + 10 = \frac{\bar{x} + 10\alpha}{\alpha}$

90. Expression = $x \cdot x^2 \cdot x^3 \ldots\ldots x^{20} \cdot \left(1 - \frac{1}{x}\right)\left(1 - \frac{2}{x^2}\right)\left(1 - \frac{3}{x^3}\right)\ldots\ldots\left(1 - \frac{20}{x^{20}}\right) = x^{210} \cdot E$

Where $E = \left(1 - \frac{1}{x}\right)\left(1 - \frac{2}{x^2}\right)\ldots\ldots\left(1 - \frac{20}{x^{20}}\right)$

Now, coefficient of x^{203} in original expression = coefficient of x^{-7} in E

But $E = 1 - \left(\frac{1}{x} + \frac{2}{x^2} + \frac{3}{x^3}\ldots\ldots\right) + \left(\frac{1}{x}\cdot\frac{6}{x^6} + \frac{2}{x^2}\cdot\frac{5}{x^5} + \frac{3}{x^3}\cdot\frac{4}{x^4} + \ldots\ldots\right) - \left(\frac{1}{x}\cdot\frac{2}{x^2}\cdot\frac{4}{x^4} + \ldots\ldots\right)$

Coefficient of $x^{-7} = -7 + 6 + 10 + 12 - 8 = 13$